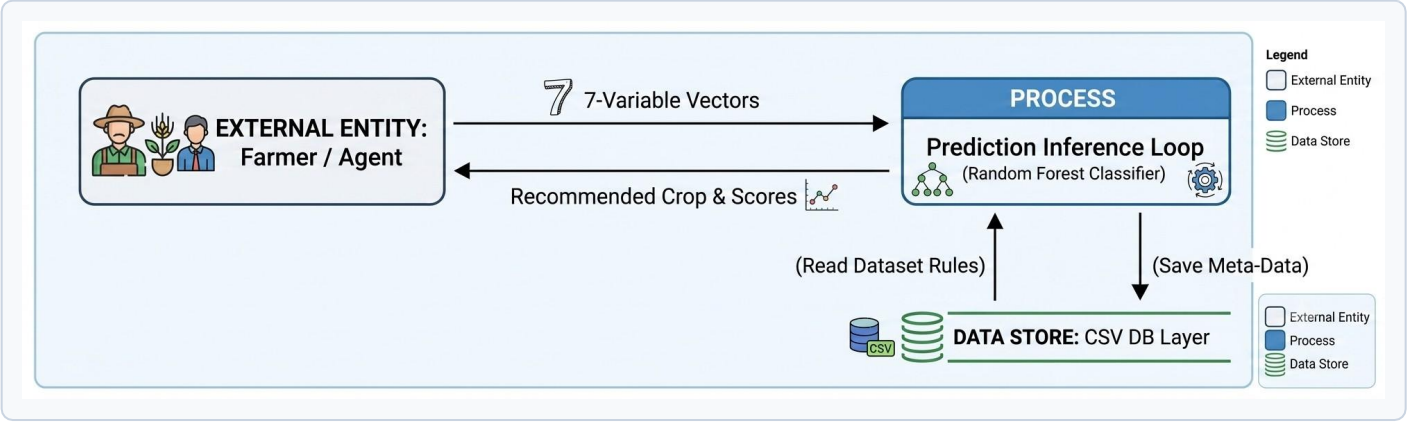
Data Flow Diagram

Date	5 Jun 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	4 Marks

Data Flow Diagrams (Level-0 Logical Architecture):

A Data Flow Diagram (DFD) visually maps out how data transitions through external entities, processes, and structured data repositories. Below is the blueprint mapping your OptiCrop workflow:



User Stories:

Use the below matrix to track specific user execution paths for validation boundaries:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Farmer)	Smart Crop Recommendation	USN-1	As a farmer, I want to enter soil and climate parameters so that the system recommends the most suitable crop for maximum yield.	Recommendation loads accurately in under 2 seconds.	High	Sprint-1
Customer (Farmer)	Crop Suitability Assessment	USN-2	As a farmer, I want to check my current soil parameters against a particular crop to evaluate its productivity compatibility potential.	Displays dynamic percentage scores clearly.	High	Sprint-1
Agricultural Researcher	Regional Data Analytics	USN-3	As an agricultural researcher, I want to analyze crop-environment distributions so that I can discover micro-farming productivity trends.	Provides clear aggregated parameter table views.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Policymaker	Policy Planning Matrix	USN-4	As a policymaker, I want to simulate critical threshold limits of inputs so that I can design sustainable regional fertilizer strategies.	Calculates deviation against system baseline means.	Medium	Sprint-2